

ANALYSIS OF GDP PER CAPITA AND LIFE EXPECTANCY

LEE LING JUN

CONTENT

INTRODUCTION

- Background motivation
- Research Question & Objectives
- Analysis Scope

DATASET OVERVIEW

- Dataset Source & Accessibilities
- Key Variables and Structure

PRE-PROCESSING

- Merging data sets
- Cleaning and Transformation of variables.
- Handling Missing values and outliers.
- Engineered Features

EXPLORATORY DATA ANALYSIS

- Descriptive Statistics
- Missing Data
- Data Quality

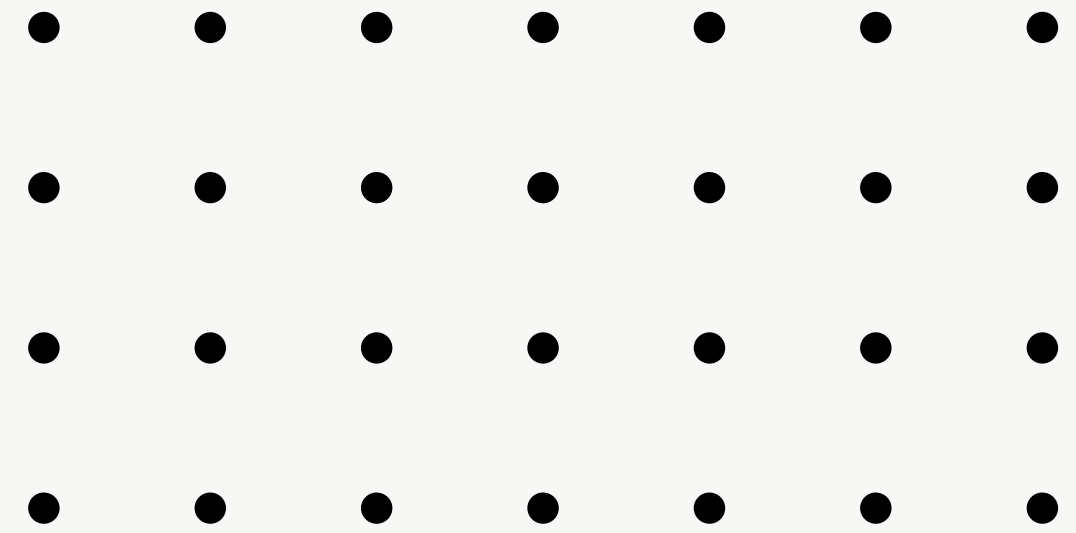
ANALYSIS & FINDINGS

- Trends Income and Longevity
- Cross-Sectional Relationship Between GDP and Life Expectancy
- Life Expectancy Differences Across Income Groups
- Regression and Outliers in GDP – Life Expectancy

CONCLUSION

- Summary of findings.
- Implications for Global Development
- Limitations of the Analysis
- Recommendations

BACKGROUND & MOTIVATION



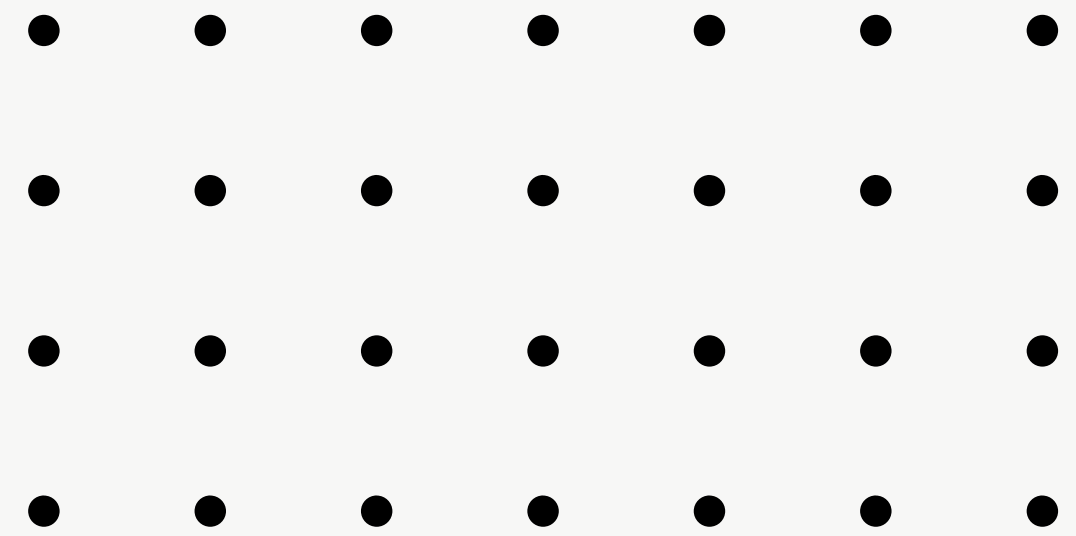
■ BROAD RISE IN ECONOMIC DEVELOPMENT AND HEALTH

- Over the past two centuries, both economic development and population health have increased.
- Higher economic development generally improves quality of life and supports longer life expectancy.

■ VARIATION DESPITE OVERALL GROWTH

- Significant differences persist: some high-income countries show only modest health outcomes, while some middle-income countries achieve unexpectedly high longevity.
- This motivates examining how closely economic prosperity and life expectancy actually move together.

RESEARCH QUESTION & OBJECTIVES



- Describe long-run global trends in both GDP per capita and life expectancy.
- Examine the cross-sectional relationship in recent years, including whether log-transforming GDP clarifies the association.
- Compare outcomes across income groups and identify countries that perform better or worse than expected given their income level.

Dataset Overview

Dataset Overview

01 GDP per Capita

The Maddison Project Database, which has estimates of GDP per capita in constant international dollars over several centuries, The dataset can be downloaded by the general public through Our World in Data at <https://ourworldindata.org/grapher/gdp-per-capita-maddison-project-database>

02 Life Expectancy

Life expectancy data: are drawn from an Our World in Data dataset containing period life expectancy estimates at multiple ages (0, 10, 15, 25, 45, 65 and 80) by country and year <https://ourworldindata.org/grapher/life-expectancy-hmd-unwpp?country=FRA~ZAF~NGA~CHN>

GDP per capita

DATASET OVERVIEW

Feature	Description
Year	Calendar year of measurement. Dataset is annual time-series.
Entity	Country or region name (e.g. Afghanistan, Spain, United Kingdom).
Code	ISO-3 country code used to uniquely identify each Entity.
Year	Calendar year of measurement. Dataset is annual time-series.
GDP per capita	Gross Domestic Product per person, measured in constant international dollars. This represents economic output adjusted for population size and inflation, making it comparable across countries and over time.
900793-annotations	Auxiliary metadata column (mostly empty). Contains annotation flags used internally by the source organisation. Not used for modelling or analysis.

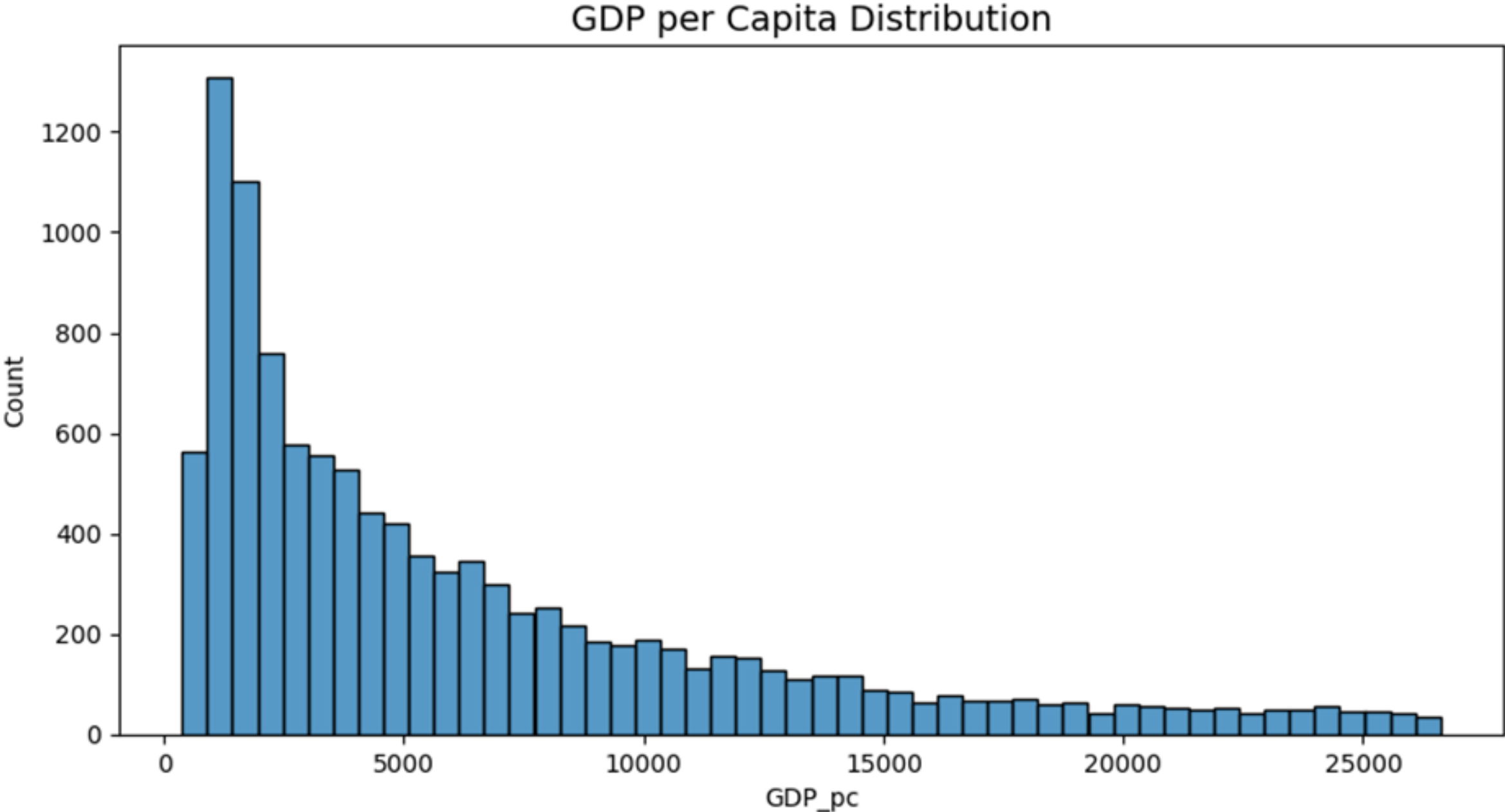
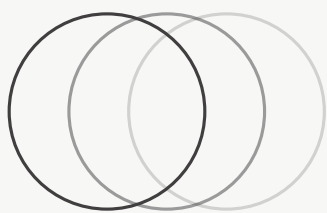
Life Expectancy

Dataset Overview

Feature	Description
Entity	Country or region name (e.g. Afghanistan)
Code	ISO-3 country code
Year	Calendar year of observation
Life expectancy variables (7 columns)	Period life expectancy measures at ages 0, 10, 15, 25, 45, 65, 80 — expected remaining years of life at each age, averaged across sexes

Descriptive Statistics:

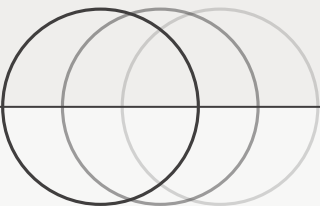
GDP per Capita



					Interquartile range (IQR)			
Feature	Count	Mean	σ	Min	25%	50%	75%	Max
Year	21,586	1,885.58	173.42	1	1,865.00	1,957.00	1,991.00	2,022.00
GDP per capita	21,586	6,869.64	10,723.73	295	1,470.00	2,617.94	7,231.69	160,051.23

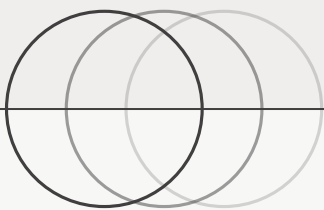
Descriptive Statistics:

Life Expectancy



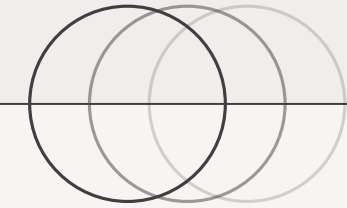
Life Expectancy								
					Interquartile range (IQR)			
Feature	Count	Mean	σ	Min	25%	50%	75%	Max
Year	20,804	1980.03	32.95	1751	1964	1984	2004	2023
Life expectancy – Age 0	20,804	62.74	12.32	<u>10.99</u>	54.18	65.19	72.23	<u>86.37</u>
Life expectancy – Age 10	20,804	59.04	7.08	10.23	53.93	59.69	64.05	76.80
Life expectancy – Age 15	20,804	54.43	6.75	6.93	49.54	54.94	59.17	71.82
Life expectancy – Age 25	20,804	45.66	5.91	4.73	41.33	45.86	49.71	61.91
Life expectancy – Age 45	20,804	28.68	4.10	3.60	25.66	28.30	31.22	42.16
Life expectancy – Age 65	20,804	13.82	2.54	3.01	11.96	13.32	15.21	23.21
Life expectancy – Age 80	20,804	6.11	1.30	2.05	5.21	5.81	6.74	11.82

Missingness:



GDP per capita dataset	
Features	Missingness (%)
900793-annotations	9.987
Code	1.26
Entity	0
Year	0
GDP per Capita	0

Life Expectancy Dataset	
Features	Missingness (%)
Code	9.3
Entity	0
Year	0
Age: 0	0
Age: 10	0
Age: 15	0
Age: 25	0
Age: 45	0
Age: 65	0
Age: 80	0



CONSISTENT MATCHING

An inner join on Entity, Code, and Year was applied to retain only observations that appear in both datasets.

This prevents artificial gaps that would occur if unmatched country–year entries were kept.

CLEAN ALIGNED PANEL STRUCTURE

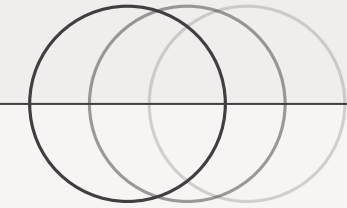
The resulting dataset contains coherent country–year rows where GDP per capita and all life-expectancy measures refer to the same observation.

This alignment ensures comparability and reliability for subsequent analysis.

Cleaning and Transformation

PREPROCESSING

G



STANDARDISATION

Long descriptive labels (e.g., “Life expectancy – Sex: total – Age: 0 – Type: period”) were shortened to clear codes like LE_0, LE_10, ... LE_80, and GDP per capita was renamed to GDP_pc to simplify coding and interpretation.

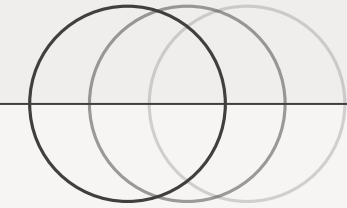
HARMONISING DATA

The Year variable was cast to integer, and all numeric fields were converted to proper numeric types, with invalid entries coerced to missing to prevent processing errors

NON ESSENTIAL

Irrelevant annotation or metadata columns were dropped so that only variables directly needed for analysis and visualisation remained, reducing clutter and improving clarity.

Handling Missing values and outliers



MISSING DATA

Columns that added no analytical value (e.g., duplicate life-expectancy measures, unused GDP variants) were removed to avoid noise in the analysis.

IRRELEVANT

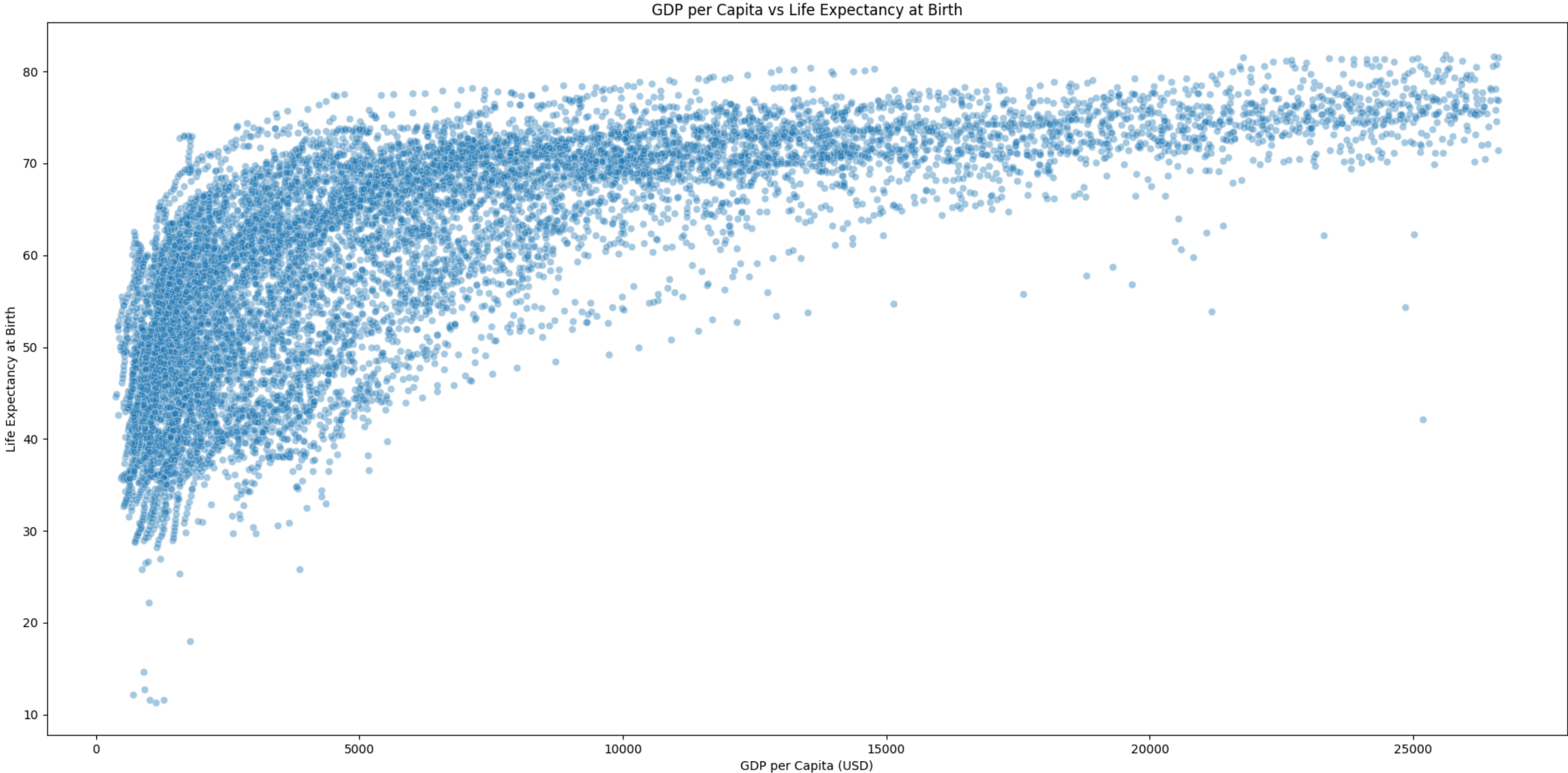
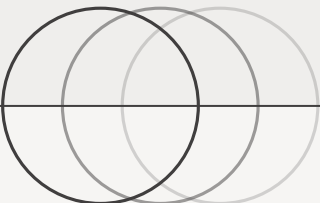
Small gaps in country time series were imputed using forward/backward fills to preserve each nation's historical trend rather than introduce artificial jumps

OUTLIER

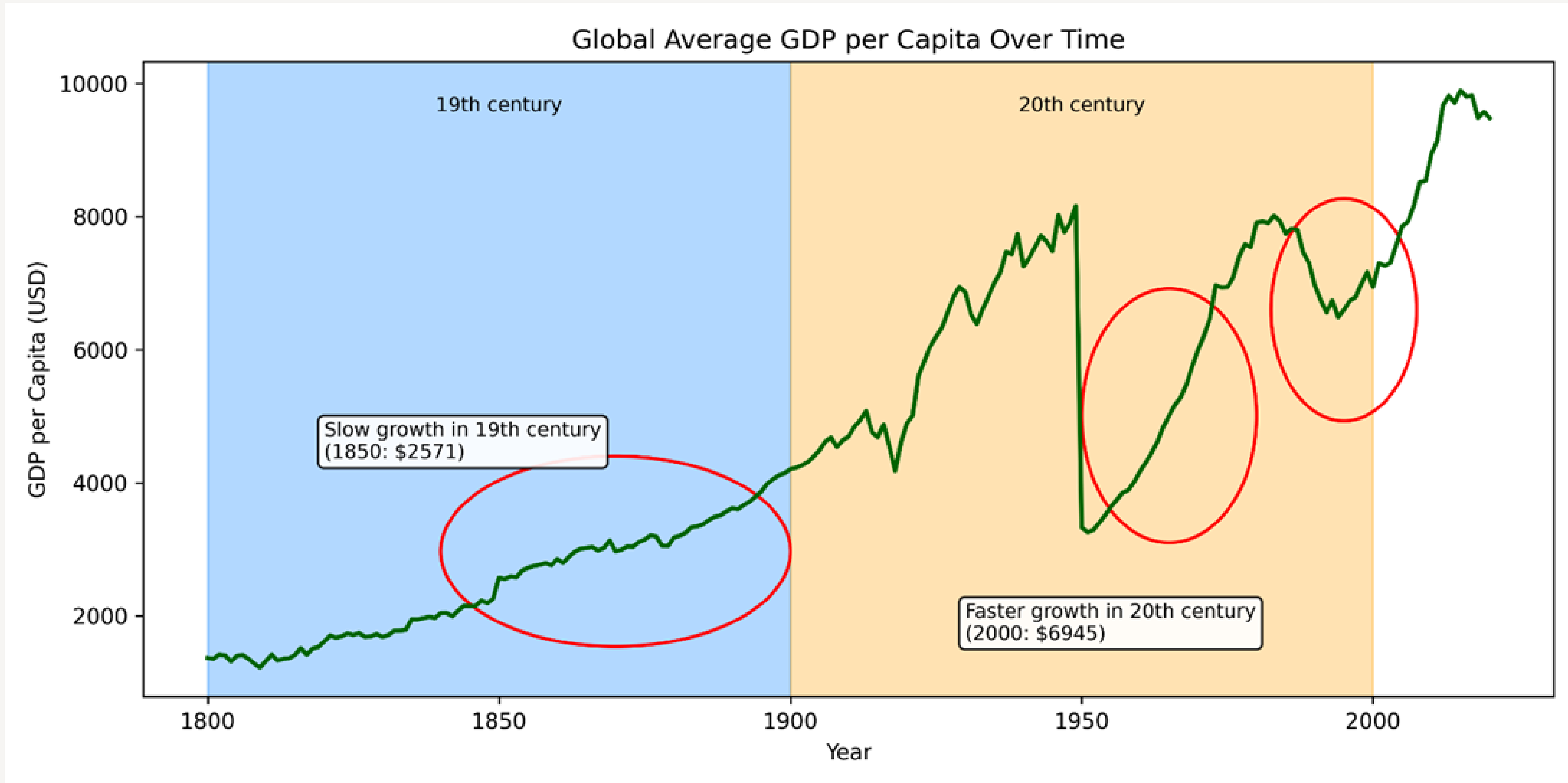
Outliers were filtered using the IQR method to reduce distortion and ensure that extreme values did not disproportionately influence the results.

Engineered Features

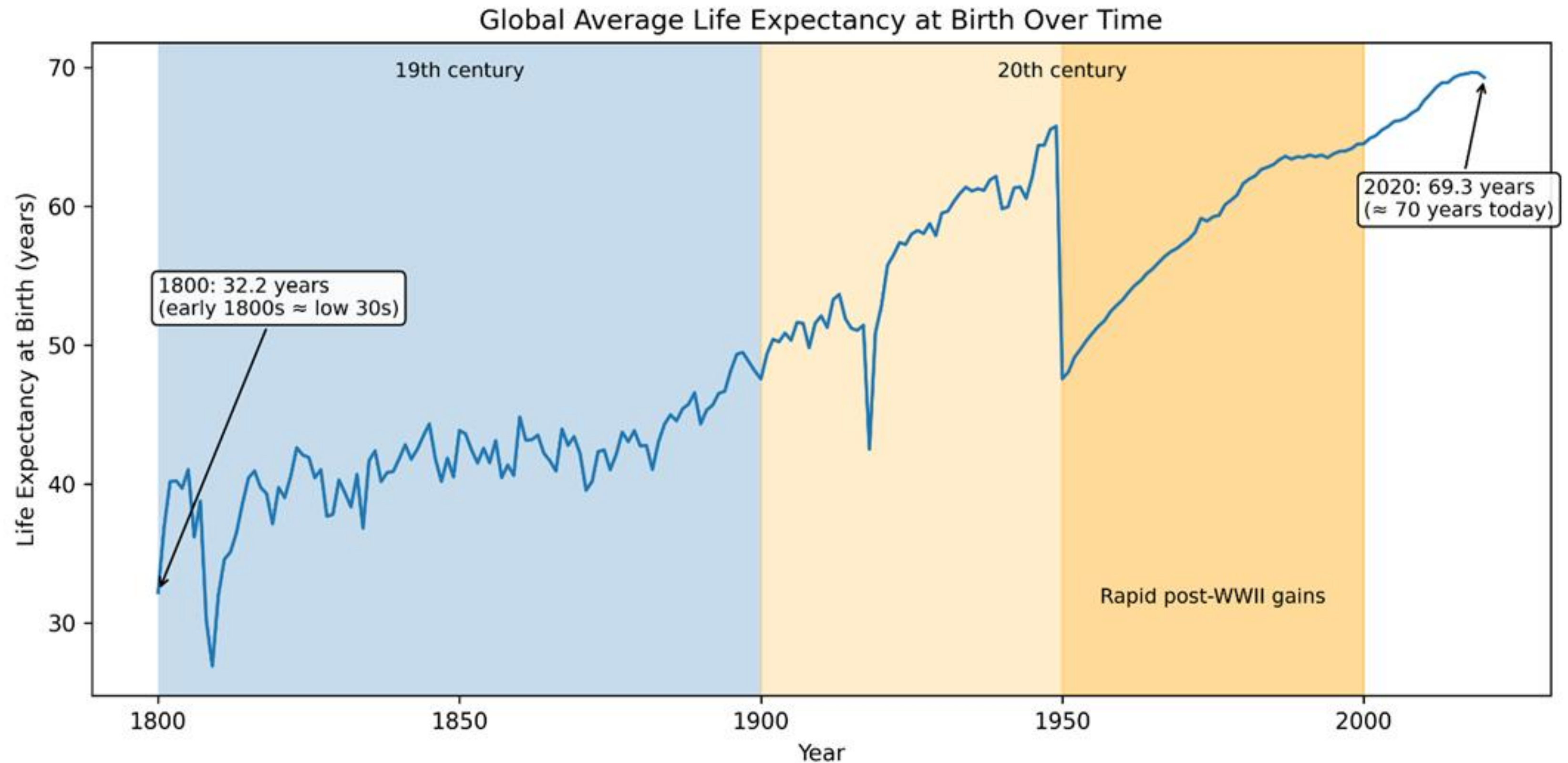
PREPROCESSING



Trends



Trends



Cross-Sectional Relationship:

Correlation Decade Mean

	GDP_pc	LE_0	LE_mean
GDP_pc	1	0.985467	0.985671
LE_0	0.985467	1	0.997781
LE_mean	0.985671	0.997781	1

Legend	
<u>GDP_pc</u>	Raw GDP per capita
<u>LE_0</u>	Life Expectancy at birth
<u>LE_mean</u>	Mean Life Expectancy across all ages

- **Very high correlations (>0.98)** across all variables.
- **GDP and life expectancy move closely together** when averaged by decade.
- **LE_0 and LE_mean almost identical** (≈ 0.998 correlation).
- Smoothing removes noise → **reveals strong long-run co-movement.**
- Shows **stable global progress** in both income and longevity over decades.

Cross-Sectional Relationship:

Correlation Pearson

VISUAL ANALYSIS

	<u>GDP_pc</u>	<u>log_GDP_pc</u>	<u>LE_0</u>	<u>LE_mean</u>
<u>GDP_pc</u>	1	0.902949	0.699438	0.714579
<u>log_GDP_pc</u>	0.902949	1	0.77806	0.772722
<u>LE_0</u>	0.699438	0.77806	1	0.980876
<u>LE_mean</u>	0.714579	0.772722	0.980876	1

<u>Legend</u>	
<u>GDP_pc</u>	Raw GDP per capita
<u>log_GDP_pc</u>	Natural Log of GDP per Capita
<u>LE_0</u>	Life Expectancy at birth
<u>LE_mean</u>	Mean Life Expectancy across all ages

- **GDP and LE₀ are strongly linked**
 - Moderate–strong positive correlation (~0.70).
 - Richer countries generally live longer.
- **Log(GDP) fits the relationship better**
 - log_GDP_pc correlates more strongly with LE₀ (~0.78).
 - Supports diminishing returns to income.
- **Life expectancy measures move together**
 - LE₀ and LE_mean correlate extremely highly (~0.98).
 - Better infant survival → better longevity at all ages.
- **GDP variables correlate by design**
 - GDP_pc and log_GDP_pc ~0.90.
 - Log is a scaling transformation of the same variable.
- **Correlation ≠ causation**
 - Income helps, but health systems, education, and stability also matter.

Cross-Sectional Relationship:

Correlation Spearman

VISUAL ANALYSIS

	GDP_pc	log_GDP_pc	LE_0	LE_mean
GDP_pc	1	1	0.804817	0.790763
log_GDP_pc	1	1	0.804817	0.790763
LE_0	0.804817	0.804817	1	0.990084
LE_mean	0.790763	0.790763	0.990084	1

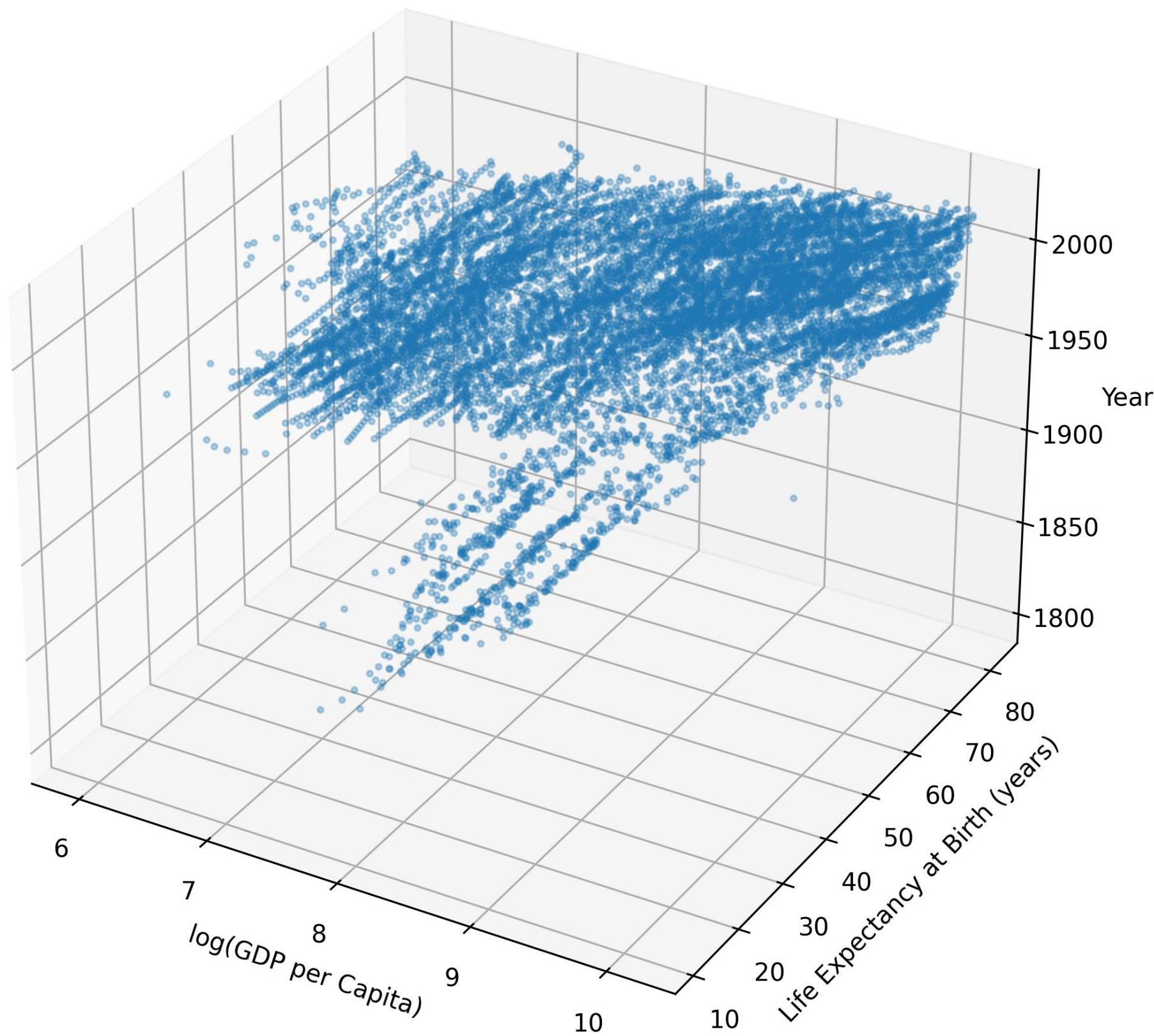
Legend	
GDP_pc	Raw GDP per capita
log_GDP_pc	Natural Log of GDP per Capita
LE_0	Life Expectancy at birth
LE_mean	Mean Life Expectancy across all ages

1. Strong monotonic link
 - Higher-income countries consistently rank higher in life expectancy. (~0.80).
2. Log(GDP) aligns well:
log_GDP_pc–LE₀ correlation is identical (~0.80).
Reinforces that *relative* income differences track life expectancy well.
3. Life expectancy measures move together
LE₀ ≈ LE_{mean} (≈0.99).
4. GDP variables rank identically:
 - GDP_pc & log_GDP_pc have perfect monotonic correlation (=1).
 - Log transform changes scale, not ordering.
5. Spearman confirms robustness
 - Not driven by outliers; trend holds in ranks.

Cross-Sectional Relationship:

VISUAL ANALYSIS

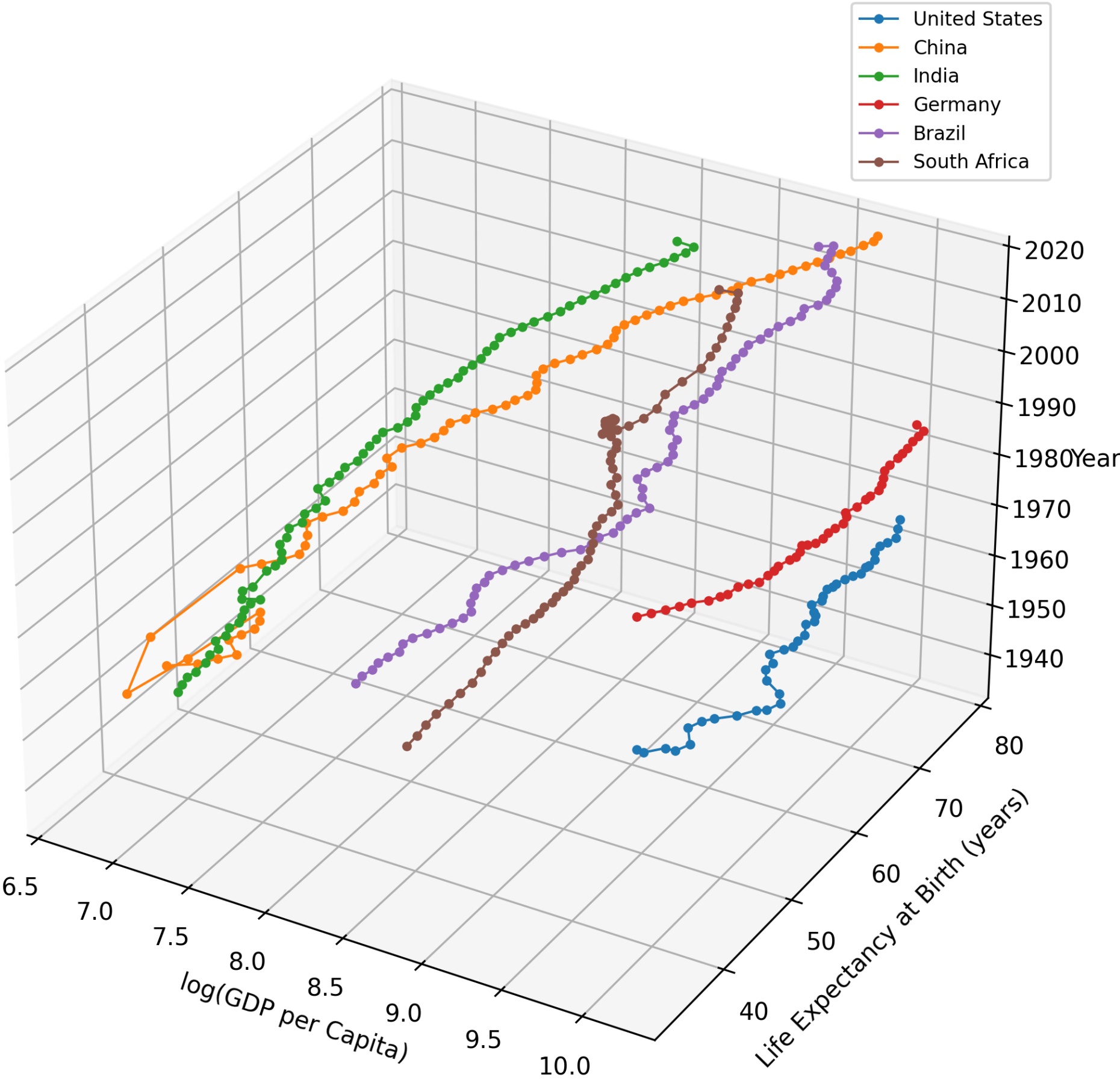
3D: log(GDP), Life Expectancy, and Time



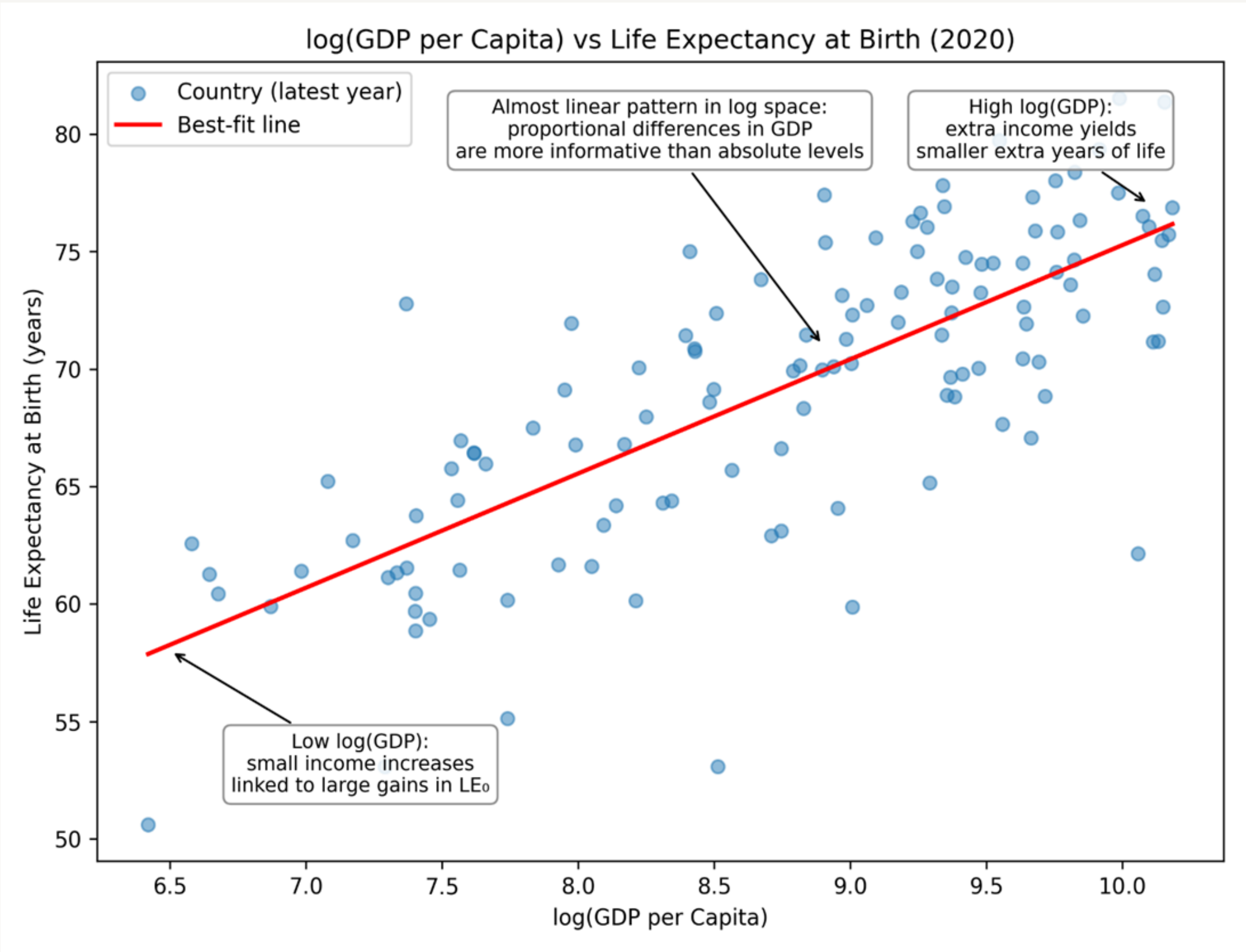
Cross-Sectional Relationship:

VISUAL ANALYSIS

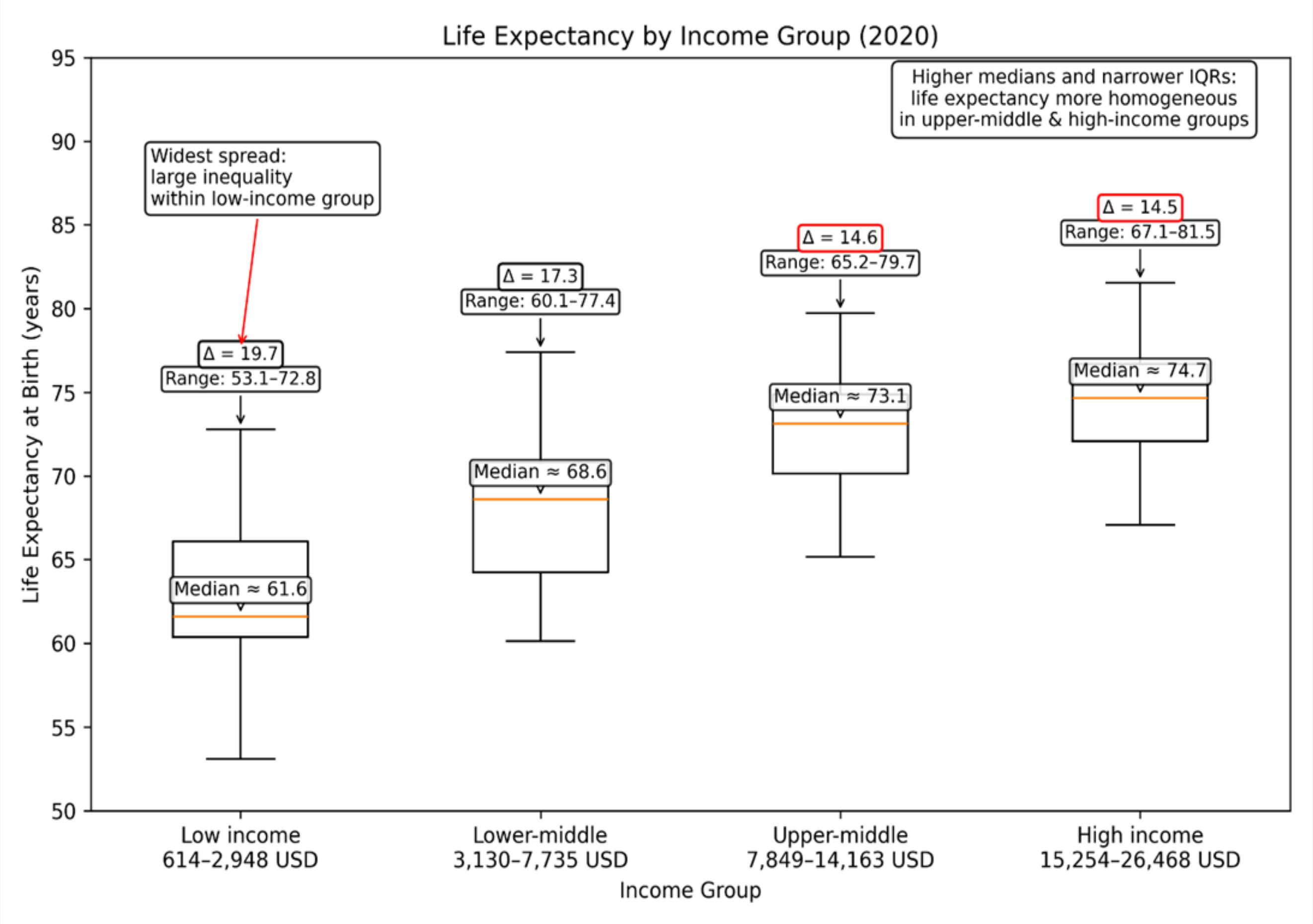
3D Trajectories: log(GDP) vs Life Expectancy vs Time



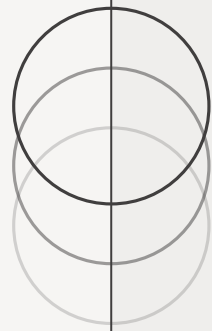
Cross-Sectional Relationship:



Cross-Sectional Relationship:



Cross-Sectional Relationship:



Summary of findings

CONCLUSION

LONG-RUN GLOBAL TRENDS

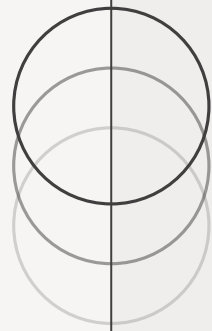
- Life expectancy: slow 19th-century rise → sharp 20th-century jump.
- GDP per capita: gradual early growth → rapid mid-20th-century increase.
- Both rise together over time.

NON-LINEAR RELATIONSHIP

- Positive but non-linear link.
- Low-income: big health gains from small GDP increases.
- High-income: small health gains despite large GDP growth.

INCOME GROUP DIFFERENCES

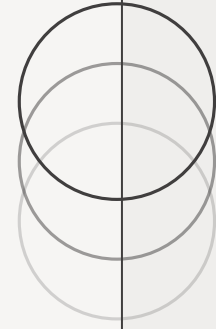
- Low-income: lowest medians, widest spread → unequal health access.
- Upper-middle/high-income: higher medians, narrow spread → more uniform conditions.



Limitation of Analysis

CONCLUSION

- Historical data, especially before the 20th century, rely on reconstructed estimates and may be incomplete or inaccurate.
- GDP per capita and life expectancy capture only broad trends and exclude deeper dimensions of wellbeing such as inequality, morbidity, and quality of life.
- The analysis is descriptive and does not account for underlying drivers (e.g., education, health systems).
- Income-group classifications simplify a complex reality, as similar GDP levels can mask very different health outcomes.



THANKYOU